SDT Module 1 — Exam Reference

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1.1 Concept of Design Thinking

Design Thinking is a **human-centered, iterative problem-solving approach**. Instead of jumping to code, you first understand the user, then design.

Key shift: "What can we build?" → "What do people actually need?"

3 Spaces of Innovation

Space	What it means	How
Inspiration	Understand the problem	Watch what users DO, listen to what they SAY, feel what they FEEL
Ideation	Generate solutions	Quantity over quality — one problem → many solutions
Implementation	Take solution to market	Iterative path from project room to product

These are **not linear** — you loop back constantly.

10 Tools: Approach to Design

#	Tool	One-line definition
1	**Visualization**	Use imagery/diagrams to envision possibilities and bring ideas to life
2	**Journey Mapping**	Track customer's end-to-end experience through their own eyes
3	**Value Chain Analysis**	Examine internal operations that support the customer journey
4	**Mind Mapping**	Synthesize research insights visually to form design criteria

#	Tool	One-line definition
5	**Brainstorming**	Generate new possibilities and alternative solutions
6	**Concept Development**	Assemble ideas into a coherent solution for evaluation
7	**Assumption Testing**	Isolate and test key assumptions that make or break a concept
8	**Rapid Prototyping**	Express concept in tangible form for testing and refinement
9	**Customer Co-Creation**	Enroll customers to directly participate in creating the solution
10	**Learning Launch**	Create an affordable real-world experiment to test with actual market data

4 PMA Tools (Project Management Aids)

PMA	Tool	Purpose
PMA 1	**Design Brief**	Formalizes project, defines goals/resources/timelines — the "North Star"
PMA 2	**Design Criteria**	Sets criteria to evaluate alternative designs (from user needs + business requirements)
PMA 3	**Napkin Pitch**	Crystalizes top 3–5 solutions in a simple comparable template
PMA 4	**Learning Guide**	Defines affordable resources to test if top concepts are feasible

1.2 The 5 Stages of Design Thinking

Process: Empathize → Define → Ideate → Prototype → Test

Key rule: Non-linear and iterative — you can return to any stage at any time.

Stage 1: Empathize

Goal: Understand the user deeply — not what they say they want, but what they actually need.

Methods:

- **Observation & Shadowing** — watch users in their natural environment to spot pain points
- **Deep Dive Interviews** — keep asking "why?" to uncover latent needs and motivations
- **Immersion** — experience the user's journey yourself, feel the frustrations firsthand

> One-liner: *Empathy is the foundation of human-centered design — setting aside assumptions to gain deep insights.*

Stage 2: Define

Goal: Frame the exact problem clearly.

- Synthesize all findings from the Empathize stage
- Create a **POV (Point of View)** statement — human-centered, not business-focused
- Use **"How Might We" (HMW)** questions to open up solution space

> Example POV: *"Students need a faster way to access exam schedules."* Example HMW: *"How might we help busy office workers eat healthier at lunch?"*

Stage 3: Ideate

Goal: Generate as many solutions as possible — quantity over quality.

Rules:

- Defer judgment
- Encourage wild ideas
- Build on others' ideas

Techniques:

- **Brainstorming** — verbal, collaborative, high energy, immediate feedback
- **Brainwriting** — write ideas silently on sticky notes; eliminates groupthink
- **Mind Mapping** — visual diagram with central idea + branches

- **SCAMPER** — Substitute, Combine, Adapt, Modify, Put to another use, Eliminate, Reverse

Stage 4: Prototype

Goal: Make ideas tangible — build to learn, not to impress.

- Low-fidelity = sketches, paper models, wireframes
- Motto: **"Fail fast to learn faster"**
- Just needs to be something users can interact with

Stage 5: Test

Goal: Put prototype in front of real users, collect feedback.

- Observe interactions and watch for friction points
- About **validating**, not selling
- Insights here often send you back to earlier stages

Diverge → Converge Pattern (important concept)

Phase	What you do	Mindset
Diverge	Generate many options	Quantity over quality
Converge	Narrow to the best one	Quality over quantity

Every stage has both — expand first, then narrow.

Stage	Diverge	Converge
Empathize	Interview many users	Build focused empathy maps/personas
Define	Write multiple HMW questions	Select single most actionable problem statement
Ideate	Brainstorm all ideas	Pick top ideas to prototype
Prototype	Build multiple low-fi versions	Choose the one that best tests the risk
Test	Collect broad feedback	Decide what to iterate, pivot, or ship

5-Mark Answer Template

For each stage write: (1) stage name + one-line goal, (2) 2–3 activities, (3) small example. Repeat for all 5.

1.3 Human-Centered Design (HCD)

Definition: Shifts focus from technology-first ("What can we build?") → people-first ("What do users actually need?"). Software adapts to humans, not the other way around.

4 Core Principles

Principle	What it means
Focus on People	Design with deep empathy for actual end-users — their frustrations, environment, workflows
Fix the Root Problem	Don't build what users SAY they want — observe to find the REAL underlying problem
Everything is a System	Software doesn't exist alone — device, internet speed, surroundings all affect experience
Iterate Continuously	First design won't be perfect — test early, fail fast, refine based on real feedback

6 Key Characteristics

- User-focused
- Empathy-driven
- Iterative
- Evidence-based
- Accessible
- Collaborative

Principles of HCSD (for 10-mark answers)

- **Understand users** — interviews, surveys, observation, usability studies
- **Involve users throughout** — planning, design, development AND testing
- **Focus on usability** — easy to learn, efficient, minimizes errors

- **Iterative design** — continuously improve based on feedback
- **Accessibility & inclusivity** — design for different abilities, languages, backgrounds

Merits of HCD

- Improves user satisfaction
- Increases productivity and efficiency
- Reduces user errors and training needed
- Enhances accessibility and inclusiveness
- **Lowers development costs** by catching problems early
- Increases adoption and customer loyalty

5-Mark Answer Template

"Human-Centered Software Design (HCD) is an approach that prioritizes the needs, goals, abilities and experiences of users. Instead of starting with technology, it begins by understanding users and designing solutions that are useful, usable and accessible." → then list the 4 core principles with one line each.

1.4 Design vs Traditional Problem Solving

Main Comparison Table

Feature	Design Thinking	Traditional Problem Solving
Focus	User-centered	Technology-centered
Process	Iterative (non-linear)	Sequential (linear)
Approach	Understand users first	Implement requirements directly
Attitude to failure	Fail fast, learn fast	Failure = bad
Feedback	Continuous throughout	Mainly after development
Experimentation	Encouraged	Follows predefined plans
Problem framing	Reframe the problem	Accept problem as given
Team	Interdisciplinary	Usually only technical team

Wicked Problems (PYP asked this twice!)

Definition: Problems that are complex, ambiguous, have no single correct solution and change as you try to solve them.

Characteristics:

- No definitive formulation
- No stopping rule — never fully "solved"
- Solutions are not right or wrong — just better or worse
- Every wicked problem is unique
- Solving it changes the problem itself

Example: Designing a healthcare app for rural users — as you solve affordability, you discover literacy issues, then connectivity issues — it keeps evolving.

Why design thinking handles wicked problems better: It's iterative, empathy-driven, and doesn't assume the problem is fixed.

The 4-Stage Design Process

Stage	Question	Phase
What is?	Understand current reality	Diverge + Converge
What if?	Visualize new future	Diverge + Converge
What wows?	Make choices for better solution	Diverge + Converge
What works?	Find feasible solution	Diverge + Converge

Crystal Ball vs Reactive Approach

Feature	Crystal Ball	Reactive/Real-time
Data used	Qualitative trends, future signals	Live data, A/B tests, crash logs
Product lifecycle	Long, strategic R&D	Fast, continuous releases
Risk	High risk, high reward	Low risk, low reward
Example	Automotive concept cars	Social media platforms

5-Mark Answer Template

"Design thinking differs from traditional problem solving in its fundamental approach. While traditional methods are technology-centered and sequential, design thinking is human-centered and iterative..." → then give the comparison table with 5–6 rows.

1.5 Case Studies

Exam tip: The question gives you a scenario and says "apply design thinking to X". Just map all 5 stages to that scenario with specific details.

IRCTC (Direct PYP Question)

Stage	What you do	Specific finding/output
Empathize	Interview passengers, ticket collectors, station staff	Users struggle with seat availability, confusing UI, slow booking at peak hours
Define	Synthesize findings into POV	POV: "Passengers need a fast, reliable way to book tickets without confusion"
Ideate	Brainstorm solutions	Real-time seat map, one-click booking, AI chatbot, offline mode, PNR push notifications
Prototype	Build low-fi wireframe	3-step booking flow, simplified seat selection layout
Test	Users test prototype	Feedback: add regional language, show platform number, simplify cancellation → iterate

Smart Campus Navigation

Stage	Application
Empathize	Interview students, visitors, faculty → students get lost first semester, maps are confusing

Stage	Application
Define	"New students need a simple interactive campus navigation system"
Ideate	GPS navigation, QR codes at entrances, AI chatbot, voice directions
Prototype	Mobile app with campus map, classroom locator, cafeteria finder
Test	Feedback → add wheelchair routes, show walking time, add classroom photos → iterate

Uber Ride Selection (PYP had this)

Stage	Application
Empathize	Observe how people book rides, interview drivers and riders
Define	"Users need to quickly select a ride type and see accurate arrival times"
Ideate	Ride selection panel, price comparison, live tracking
Prototype	Clickable mockup of ride selection screen
Test	Users confused by pricing → simplify and iterate

10-Mark Answer Structure

Intro (2 lines): *"Design thinking is a human-centered iterative approach. Applying it to [X] involves the following 5 stages:"*

For each stage (5 stages × ~1.5 marks each):

1. Stage name + one-line definition
2. 2–3 activities specific to that scenario
3. One concrete output or finding

Conclusion (1 line): *"This iterative process ensures the final solution is user-validated and meets real needs rather than assumed ones."*